

Safety Equipment Module

Personal Protective Equipment (PPE) — Marine Engineering Notes

1. Introduction to PPE

Personal Protective Equipment (PPE) refers to specialized clothing and equipment worn by seafarers and engine room personnel to protect themselves against health and safety hazards encountered during shipboard operations. On board a ship, engineers and crew members are regularly exposed to risks such as moving machinery, high noise levels, hot surfaces, chemicals, electrical hazards, falling objects, and slippery surfaces. PPE acts as the last line of defense against these hazards, after engineering controls and safe working procedures have been applied.

Why PPE is Important on Ships:

- Ships operate in confined, hazardous environments (engine rooms, cargo holds, deck areas).
- Reduces the risk and severity of injury during accidents.
- Mandatory requirement under SOLAS, ISM Code, and STCW guidelines.
- Protects against multiple hazards simultaneously (heat, noise, chemicals, impact).
- Non-compliance can lead to serious injury, port state control detention, or company penalties.

General Principles of PPE Use:

- PPE must be selected based on the specific hazard/task (risk assessment).
- It should fit properly and be worn correctly at all times in designated areas.
- PPE must be inspected before use and maintained/replaced when damaged.
- PPE is a supplement to, not a substitute for, safe working practices.
- Training must be provided on correct use, limitations, and storage of PPE.

2. Safety Helmet

The safety helmet (hard hat) protects the head from falling objects, low overhead obstructions, and impact injuries, which are common in engine rooms and on deck due to overhead pipework, cranes, and suspended loads.

- Made of high-impact resistant plastic (HDPE) shell with an adjustable inner harness.
- Protects against falling tools, impact with fixed structures, and electrical hazards (Class E types).
- Should be replaced if cracked, dented, or after a significant impact.
- Chin strap should be used in windy or high areas to prevent it from falling off.
- Must be worn in engine rooms, cargo holds, and during mooring/anchoring operations.

3. Safety Glasses / Goggles

Safety glasses and goggles protect the eyes from flying particles, chemical splashes, dust, and harmful UV/IR radiation generated during welding, grinding, or chemical handling tasks.

- Impact-resistant lenses protect against flying debris during grinding, chipping, or drilling.
- Goggles with sealed edges are used when handling chemicals or during battery maintenance.
- Welding goggles/shields have tinted lenses to protect against arc flash and UV radiation.
- Should be cleaned regularly and stored in a case to avoid scratches.
- Prescription-glasses wearers should use over-goggles or prescription safety glasses.

4. Safety Ear Muffs / Ear Plugs

Hearing protection is essential in the engine room, where noise levels from main engines, generators, and machinery frequently exceed safe exposure limits (often above 100 dB).

- Ear muffs cover the entire outer ear and offer high noise reduction, ideal for engine room duty.
- Ear plugs are inserted into the ear canal, useful for quick tasks and use under helmets.
- Prolonged unprotected exposure to high noise levels can cause permanent hearing loss.
- Noise Reduction Rating (NRR) should match the noise level of the working area.
- Must be worn in designated high-noise areas marked with hearing-protection signage.

5. Safety Gloves

Gloves protect the hands from cuts, burns, chemical exposure, and electrical hazards. The correct type of glove must be selected according to the task being performed.

- Cotton/leather gloves — general mechanical work, handling tools and rough surfaces.
- Rubber/neoprene gloves — protection against chemicals, acids, and battery maintenance.
- Insulated gloves — protection during electrical work to prevent shock.
- Heat-resistant gloves — used when handling hot machinery parts or during welding.
- Gloves must be inspected for tears or punctures before use and replaced when damaged.
- Gloves should not be worn near rotating machinery due to entanglement risk.

6. Safety Boots

Safety boots protect the feet from falling objects, punctures, slips, and electrical hazards, and provide grip on oily or wet deck and engine room surfaces.

- Steel-toe or composite-toe cap protects against crushing injuries from falling objects.
- Slip-resistant, oil-resistant soles reduce the risk of slips in engine rooms and on deck.
- Puncture-resistant midsoles guard against sharp objects like nails or metal shards.
- Should be well-fitted, laced properly, and kept dry to prevent fungal infections.
- Mandatory footwear at all times when on duty in machinery spaces and open decks.

7. Safety Coverall

The boiler suit or coverall protects the body and regular clothing from oil, grease, heat, and mechanical hazards while working in the engine room or on deck.

- Made from durable, often flame-retardant fabric for protection against sparks and heat.
- Should fit properly — not too loose (entanglement risk) and not too tight (restricts movement).
- Long sleeves and full-length legs protect skin from oil, chemical splashes, and abrasions.

- Should be kept clean; oil-soaked coveralls increase fire risk and skin irritation.
- High-visibility coveralls/vests may be required for deck work and low-light conditions.

8. Quick Reference Summary

PPE Item	Main Hazard Protected Against
Safety Helmet	Falling objects, head impact
Safety Glasses/Goggles	Flying particles, chemical splash, UV/arc light
Ear Muffs/Plugs	High noise levels (hearing damage)
Safety Gloves	Cuts, burns, chemicals, electric shock
Safety Boots	Crush injury, punctures, slips
Safety Coverall	Oil/grease, heat, sparks, abrasion

Note: PPE must always be used in combination with proper training, risk assessment, and adherence to the ship's Safety Management System (SMS) as per the ISM Code.